

“UG” END SEMESTER EXAMINATION – MARCH, 2021
(OPEN ELECTIVE - REGULAR)
3RD YEAR (5TH SEMESTER)

STRENGTH OF MATERIALS

MENUGOE-03

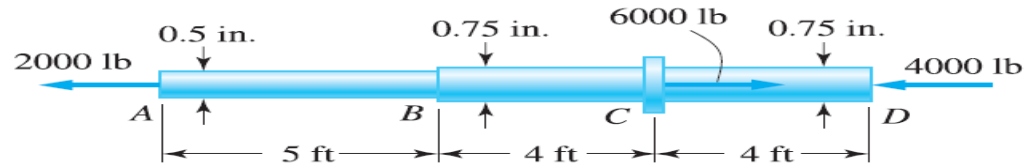
[Full Marks: 80]

[Time: 3hrs.]

(Attempt any Eight)

8x10

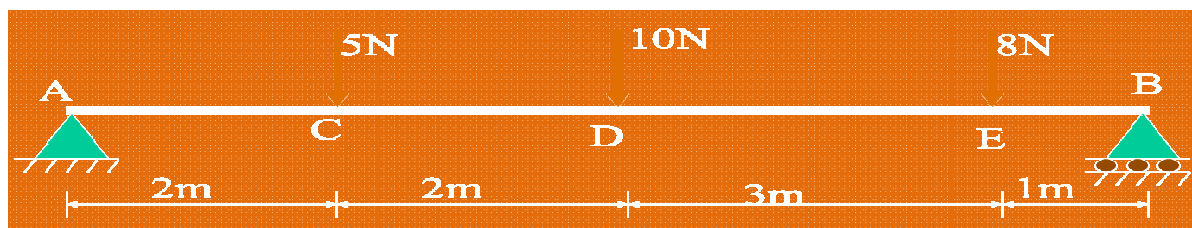
1. Determine the total elongation or deformation in the bar.



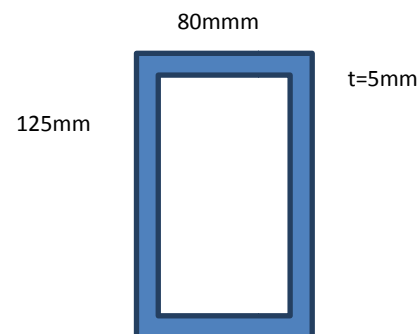
2. A) what is stress concentration?

B) A rigid wheel is 3 meters in diameter. It is desired to shrink on to the wheel a thin steel tyre. Find the internal diameter of the tyre if after fitting the hoops stress in the tyre is 90 N/mm^2 . Find also the least temperature to which the tyre must be heated above that of the wheel. (Take $E=2 \times 10^5 \text{ N/mm}^2$ and $\alpha=1.2 \times 10^{-5} \text{ per } ^\circ\text{C}$)

3. Draw shear force and bending moment diagrams [SFD and BMD] for a simply supported beam subjected to three point loads as shown in the Fig. given below.



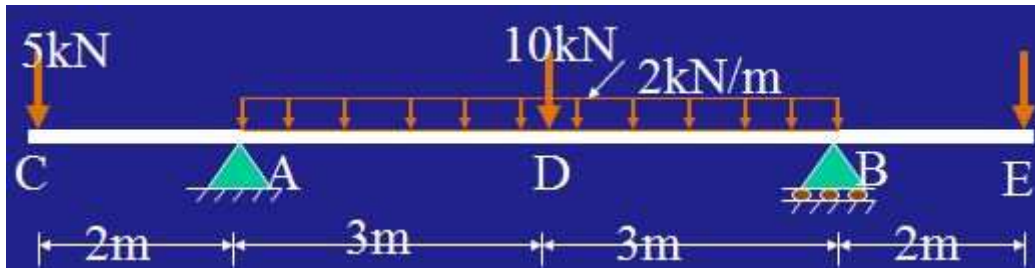
4. Fig. shows a rectangular tube section of an aluminium alloy whose ultimate stress is 405 N/mm^2 . Determine the bending moment for which the factor of safety will be 3. Find also the corresponding radius of curvature. Take $E=7 \times 10^4 \text{ N/mm}^2$.



5. A cantilever AB of length l carries three point loads, W each at distances $\frac{l}{3}$, $\frac{2l}{3}$ and l from the fixed end A. Determine the slope and deflection at the free end B.

6. A cantilever of length l is subjected to a couple M at its free end. Find the slope and deflection of the end.

7. Draw SFD and BMD for the double side overhanging beam subjected to loading as shown below. Locate points of contraflexure if any.



8. (a) The modulus of elasticity and rigidity of a material are 200 GPa and 80 GPa respectively. Find the Poisson's ratio and Bulk modulus.

(b) A rod meant to be positioned in a framework is 1800 mm long but is found to be 1.5 mm too short. In order to fit the member in the framework, the rod is heated uniformly to enable it to be fitted. Find the necessary rise in temperature needed. What would be the stress in the rod as it cools back to the original temperature if its end is prevented from movement? Take $E = 2 \times 10^5 \text{ N/mm}^2$ and $\alpha = 0.000012 \text{ per } ^\circ\text{C}$.

9. A bar consists of three lengths l_1, l_2, l_3 and diameter d_1, d_2, d_3 and is subjected to an axial pull P . Show that the extension of the rod is given by
$$\Delta l = \frac{4P}{\pi E} \left(\frac{l}{d_1^2} + \frac{l}{d_2^2} + \frac{l}{d_3^2} \right)$$

===== END =====

Aliah University

End Term Examination (Odd Semester), 2020-21

Course: Civil Engineering; Year: 3rd Year; Subject: Transportation Engg.-I

Subject Code: CE 307 Max. Marks: 80 Times: 3 hrs

Answer all questions

Q.1 a) Write short notes on: (i) Social Effect of Transportation (ii) Historical Development of Road Construction

b) Briefly mention the salient feature of Second 20-year Road Plan. **(12+8)**

Q. 2. a) What are the significant recommendations of Jayakar Committee report? Mention how this helped in road development in India.

b) What are the conditions which necessitate taking up of a re-alignment project of the highway?

(10+10)

Q.3. a) Derive an expression for finding the extra widening on horizontal curve.

b) Find the safe overtaking sight distance for a highway having a design speed of 120 km/h. Assume all data suitably. **(8+12)**

Q.4 Explain transition curve. What are objectives of providing transition curve?

Calculate the length of the transition curve and shift using the following data;

- ✓ Design speed= 65 kmph
- ✓ Radius of circular curve= 220 m
- ✓ Allowable rate of superelevation= 1 in 150
- ✓ Pavement rotated about the centre line of the pavement
- ✓ Pavement width including extra widening= 7.5 m

(8+12)

ALIAH UNIVERSITY
End-Term Examination- 5th Semester
Session-2020-2021

Year: 3rd Year
Semester: 5th Semester
Subject: Soil Mechanics
Subject Code: CE305

Full Marks: 80
Time: 3hr.

Answer question no.1 and any three from the rest:

1. Answer the question briefly

[2*10=20]

- i) What is porosity and void ratio?
- ii) What is flow index and liquidity index?
- iii) What is the numerical value of I_p if the soil is not plastic?
- iv) What are the main differences of compaction and consolidation?
- v) Define the terms relative density and activity.
- vi) Write the definition of residual soil and transported soil.
- vii) What is flow net?
- viii) What should be the ranges of GI for a good and poor subgrade material?
- ix) Why deflocculating agent is used in hydrometer test?
- x) Write the name of two in-situ method of unit weight determination?

2. a) Give the grain size ranges of different soil types according to IS:1498(1970). [3]

b) What are the significant of C_u and C_c in soil grain size distribution of soil? [3]

c) Define Atterberg's consistency limits with illustration. [4]

d) A soil sample has void ratio of 0.72, moisture content = 12% and $G_s=2.72$

determine the following:

a) Dry unit weight, moist unit weight (KN/m³).

b) Weight of water in KN/m³ to be added for 80% degree of saturation.

c) Is it possible to reach a water content of 30% without change the present void ratio? [10]

3. a) Neatly draw the vertical stress distribution diagram based on Boussinesq equation. What is isobar and pressure diagram? [5+5]

b) Explain Darcy's law of fluid flow through porous medium. Show that seepage velocity is always greater than superficial velocity or discharge velocity. [5+5]

4. a) Prove that for a sedimentary (layered) soil deposit, permeability is always greater for flow in horizontal direction than in the vertical direction. [12]

b) What is 'silica tetrahedron' and an 'alumina octahedron'? How are silica sheet and alumina sheet formed? Show their schematic diagram. [4+4]

5. a) Illustrate the hydrodynamic condition when effective stress will be increased. [3]
- b) Explain the shape of compaction curve in a cohesive soil in terms of the development of water films around the particles. Why the shape of compaction curve for sand is in reverse direction initially? [4+4]
- c) What is effective stress? [1]
- d) What is quick sand condition? [3]
- e) An undisturbed sample of clayey soil is found to have a wet weight of 285 N, a dry weight of 250 N, and a total volume of $14 \times 10^3 \text{ cm}^3$ if the specific gravity of soil solids is 2.70, determine the water content, void ratio and the degree of saturation. [5]

ALIAH UNIVERSITY

End-Semester Examination- March, 2021

Year: 3rd Year (Dept. of Civil Engg.)

Semester: 5th Semester

Subject: Open Channel Flow

Subject Code: CE 303

Full Marks: 80

Time: 3 hr.

Attempt any **five** questions and assume suitable data if required.

1. (a) Discuss the following terms as: 8
 - I. Open channel flow and pipe flow.
 - II. Prismatic, non prismatic channel, rigid and mobile boundary channel.
 - III. Steady-unsteady flow, Uniform-non uniform flow, gradually and rapidly flows.
 - IV. Velocity distribution and pressure distribution on linear and curvilinear flow.

(b) The velocity distribution in a channel section may be approximated by the equation, $V = V_o(y/y_o)^n$, in which V is the flow velocity at depth y ; V_o is the flow velocity at depth y_o , and $n = a$ constant. Derive expressions for the energy and momentum coefficients. 8
2. (a) A 5.0 m wide rectangular channel carries a discharge of $6.4 \text{ m}^3/\text{s}$ at a depth of 0.8 m. At a section there is a smooth drop of 0.22 m in the bed. What is the water surface elevation downstream of the drop? 7

(b) Find the discharge in the following channel section with a bed slope of 0.0006 and $n = 0.016$: 9
 - I) Rectangular, $B = 3.0 \text{ m}$, $y_o = 1.2 \text{ m}$
 - II) Trapezoidal, $B = 3.0 \text{ m}$, $y_o = 1.10 \text{ m}$ and $m = 1.5$
 - III) Triangular, $y_o = 1.5 \text{ m}$ and $m = 1.5$
3. (a) A trapezoidal channel of 10.0 m bed width and $m = 1.5$ carries a discharge of $15.0 \text{ m}^3/\text{s}$ at a depth of 1.0 m. Calculate (i) if the channel is lined with smooth concrete, and (ii) if the channel is unlined. 8

(b) What do you mean by equivalent roughness? Derive mathematical expression for equivalent roughness using Horton's Method. 8
4. (a) What do you mean by GVF? Discuss with suitable assumption and mathematical expression. 8

(b) Discuss the GVF classification, GVF profile and feature of GVF profiles. 8
5. (a) Discuss the various types of jump and its characteristics. 8

(b) What do you mean by hydraulically efficient channel section? Derive the relationship between the various geometric elements to form an efficient trapezoidal section. 8
6. (a) In a hydraulic jump taking place in a rectangular channel the discharge intensity and head loss are found to be $4.7 \text{ m}^3/\text{s}/\text{m}$ and 6.0 m respectively. Determine the sequent depths of the jump. 8

(b) A trapezoidal channel $B = 4.5 \text{ m}$, $m = 1.0$, $n = 0.03$ has a bed slope of $S_o = 0.003$. The channel ends in a sudden drop. Calculate the GVF profile for a discharge of $25.0 \text{ m}^3/\text{s}$. 8

Aliah University

End-Term Examination, March- 2021

Course: B. Tech. (Civil Engineering)

Year & Semester: 3rd Yr. (V Sem.)

Subject: Design of Steel Structures

Code: CE 301

Full Marks: 80

Time: 3hrs.

Answer any four questions. All questions carry equal marks. Assume suitable data if required.

1. Design a single angle to carry a tension of $200+X$ kN (where X = last two digits of your Roll No.) which is connected to a gusset plate of 16 mm thick with one line of 20 mm diameter bolts of grade 8.8. (Given $f_y = 250$ MPa, $f_u = 410$ MPa, $f_{ub} = 800$ MPa, $A_{nb}=245$ mm²)

[20]

2. (i) Why a separate provision for the design of a single angle compression member has been proposed by IS:800 code?

(ii) Determine the load carrying capacity of ISMC 250 and ISMB 250 as a column. These columns are laterally supported in the plane of major axis at the height of 4.0 m and in the plane of minor axis at the height of 3.0 m. Both the ends may be assumed as hinged.

[6+14]

3. Design a built-up column of 6.0 m effective length to carry a factored axial load of $1200+10X$ kN (where X = last two digits of your Roll No.) using single lacing.

[20]

4. (i) What do you meant by Plastic method of design? Why this method is more useful for indeterminate structures? Explain.

(ii) A beam of span 5 m carrying a distributed load of 30 kN/m (including self-weight) throughout the span. Design the beam assuming that it is laterally supported along its length.

[08+12]

5. (i) Explain different failure modes of bolted joints.

(ii) Discuss briefly welding methods, types of welds and common weld defects.

[08+12]